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# #12 Constructors in Java.

- Defaults values
  Integer → 0
  Floating → 0.0
  Boolean → false
  String → NULL

Constructor → used to create an object.
when we are constructing object we are putting some values automatically

* Rules of constructor
  ↳ 1) same name as class
  ↳ 2) No return type, not even void
  ↳ 3) Automatically called during object creation
  4) used to initialize an object
  5) It can also be overloaded

eg:

```
class student {
    String name;
    int age;
    String college;
    student C){

}
```

```
student s1 = new
student();
```

Types of constructors
1) Default constructor — created by Java
2) Parameterized constructor

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- Parameterized constructor is a special type of constructor that accepts arguments to initialize an object with specific values at time of its creation.

- Note: once you created your own parameterized or default constructor, then java do not create its default constructor internally.
But if you want default constructor, then constructor overload comes to point.

This keyword
→ Refers to current object

eg:

```
class student {
  string name;
  int age;
  int rollno;
  student (string name, int age, int rollno)
  {
    this.name = name;
    this.age = age;
  }
```

* Interview Q. can we call constructor manually?
→ NO
constructor is called only when obj is created
student s = new student ()
or
(2) using this (name, age);
keyword

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Q2. A new keyword allocates memory dynamically. so when heap memory is already filled & no space to allocate.
→ we get run-time exception, when not enough space available in heap.

* chaining in constructor
↓
constructor overload
&
chaining → used to optimize long code.

using this keyword we can call one constructor from another constructor.

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* Object Deep Dive | size, call by value V References | shallow vs Deep copy.

- Reference variable store in stack memory takes 4 bytes depend on JVM

- object size
  ↳ ① header size
  ↳ ② Exact field
  ↳ ③ padding.

- call by value
  call by Reference

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(1) call by value

- Passing a copy of the variable to a function / Method
- Any changes made inside the method do not affect the original variable.
- Java is always call by value.

eg:

```
class Demo{
    main(){
        int x = 4;
        int y = 5;
        s.o.p (x,y)
        addTen (x,y);
    }
    addTen (int x, int y)
        x = x + 10;
        y = y + 10;
}
}
```

(2) call by Reference

```
class Demo{
    main(){
        Random r1 = new Random(4,5);
        s.o.p (r1.x, r1.y)
    }   addTen
    addTen (random r1) {
}
class Random{
    int x;
    int y;
}
```

There is NO call by Reference in Java

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Deep and shallow copy

1. shallow copy
- copies only references (address)
- objects share same memory

2. Deep copy
creating a completely new object with same values, but diff memory location.